Jack Klawitter

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Education

Rutgers University — New Brunswick, NJ
Doctor of Philosophy (Ph.D.) in Computer Science (Enrolled) — *Sep. 2023 – Present*

- Advisor: Dr. Konstantinos Michmizos | Computational Brain Lab, CBIM

Middlebury College — Middlebury, VT
Bachelor of Arts with Majors in Physics and Computer Science, Magna Cum Laude — *Sep. 2018 – Feb. 2023*

Experience

Graduate Research Assistant — Sep. 2023 – Present
Rutgers University — *New Brunswick, NJ*

- Developed interpretable machine learning models for decoding high-dimensional neural recordings, enabling analysis of representational drift across motor learning
- Introduced Critical Spike Attribution, a novel feature attribution method for spiking neural networks, providing causal insight into spike-level contributions to model predictions
- Applied representation analysis and feature attribution to align model dynamics with biological change
- Led collaboration between Computer Science and Psychology labs to integrate modeling and experimental data collection

Research Assistant — Feb. 2022 – Dec. 2022
Middlebury College Physics Department — *Middlebury, VT*

- Developed Python script to correct the spectra of red quasars
- Published open-source code for use by the astrophysics community

Machine Learning Intern — May 2022 – August 2022
Rolls-Royce — *Indianapolis, IN*

- Developed deep learning models for early detection of jet engine failures, improving detection rate by approximately 10% in operational data

Undergraduate Researcher — May 2021 – July 2021
Ithaca College Dynamical Systems REU — *Ithaca, NY*

- Studied billiard dynamics on surfaces of revolution
- Developed a Mathematica notebook that models billiard dynamics on any surface of revolution within a “billiard board” — any region bounded by a piecewise-defined curve
- Developed original billiard constructions, including a new “unilluminable room” on the sphere and a novel result regarding the existence of periodic orbits on surfaces of revolution.
- Presented findings at the 2022 Joint Mathematics Meeting poster session

Research Assistant — May 2020 – August 2020
Middlebury College Math Department — *Middlebury, VT*

- Worked on Martin Gardner’s “Minimum No-3-In-A-Line” Problem
- Examined the use of SAT Solvers as they apply to combinatorial problems
- Modified programs written by Dr. Donald Knuth and created novel Python scripts

Publications

- **Klawitter, J.**, Michmizos, K. **Critical Spike Attribution for Feature Importance in Spiking Neural Networks**. *Neuro-Inspired Computational Elements (NICE)*, 2026 (Oral Presentation)
- **Klawitter, J.**, Michmizos, K. **Sequential Learning in Neural Decoders Reveals Interpretable Neural Representations**. *Neurocomputing*, 2026. 132742, ISSN 0925-2312.

Awards & Honors

Rutgers Brain Health Institute Trainee Travel Award, Spring 2026

Teaching

Course Instructor
Design and Analysis of Computer Algorithms — Summer 2024

Teaching Assistant
Intro to Discrete Structures I — Spring 2024, Spring 2025
Intro to Discrete Structures II — Fall 2024
Intro to Artificial Intelligence — Summer 2025

Skills

- **Programming:** Python (NumPy, PyTorch, scikit-learn)
- **Machine Learning:** Deep learning, spiking neural networks, neural decoding, time-series modeling, representation analysis, feature attribution, interpretable ML
- **Tools:** Git, LaTeX